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Submitter Information

Email: [redacted]
Organization: Darktrace

General Comment

See attached file(s)
Darktrace response to NSF / OSTP Request for Information on the Development of an Artificial Intelligence (AI) Action Plan

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Attachments

Darktrace response OSTP NSF RFI on the Development of an AI Action Plan

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Background

Darktrace, a global leader in AI cybersecurity, welcomes the opportunity to respond to this Request for Information.

The recently unveiled Chinese operation known as Volt Typhoon, in which PRC-sponsored cyber attackers infiltrated U.S. IT systems, exemplifies how vulnerable the U.S.'s most critical organizations – and the services on which Americans rely on everyday - are to increasingly stealthy and sophisticated cyber threats. Successful attacks like these show the importance of continually modernizing cybersecurity, to ensure the most effective, state-of-the-art defenses are employed against advanced and worsening cyber threats.

Adversarial AI makes the task of the defender harder. Indeed, AI can automate and scale up attacks, lowering the barrier to entry for attackers and making attacks more sophisticated and harder to detect. The FBI warned last year of increasing use of AI by cyber criminals, with attackers leveraging AI tools to craft highly convincing voice messages or emails. We have also seen AI being used by hostile states to enhance cyber-attacks: research by Microsoft and OpenAI identified state-affiliated actors leveraging LLMs to launch cyber-attacks, including for reconnaissance purposes and to create malware designed to avoid detection.

At the same time, AI-powered cyber defenses are emerging as the most effective response to this evolving threat landscape and combatting near-peer adversaries. AI systems can analyze vast amounts of data in real-time, identifying and mitigating threats more efficiently than traditional methods. AI allows for the intelligent augmentation of security teams to accelerate response times, increase efficiency, and reduce team workload. This proactive approach allows organizations to adopt preventative measures, managing more potential attack vectors and being better prepared to detect and respond to known, unknown, and novel attacks. The U.S. House of Representatives' Bipartisan AI Task Force's final report in 2024 found that "as sophisticated attackers seek to use AI offensively to exploit complex systems, security teams must use AI defensively to improve cybersecurity resiliency."

Importantly, adopting AI-enabled cyber defenses will be essential to ensure that the U.S. can maintain and enhance its leadership in AI. The Trump Administration has highlighted the value of AI for national security. The Action Plan should treat state-of-the-art cyber defenses, including specifically AI-powered cyber defenses, as an enabler of AI adoption and a prerequisite to protecting the U.S.' most critical organizations and services, as well as the country's broader national security.

Darktrace welcomes the OSTP's plans to introduce new policy actions aimed at maintaining and advancing America's leadership in AI. With over a decade of experience in applying AI to cybersecurity, we have drawn on our expertise to put forward three policy recommendations for the development of the U.S.' new AI Action Plan:

1. **Issue: Prioritize the use of AI in cybersecurity solutions across federal government use cases, ensuring enhanced national resilience against evolving cyber threats.**

The US Federal government should require cybersecurity solutions to leverage AI and Machine Learning.

- The cyber threat is getting worse, endangering economic and national security, the safety of citizens, and undermining the U.S.' ability to fully realize its ambition of retaining and enhancing its AI leadership.
- AI-powered cybersecurity systems act as the best defense against a worsening threat environment. AI and Machine Learning cybersecurity systems enhance the detection of known as well as unknown attacks, enable autonomous response at machine speed, and close the gap between defenders and attackers.
- The increasing convergence of IT and OT environments due to business or functional requirements has created greater vulnerabilities across U.S. critical infrastructure. Adopting AI-powered cybersecurity that can operate within and across both IT and OT environments is essential to have a unified defense posture and the only way to secure the critical resources Americans rely on every day.
- AI also helps to address the nation's cyber skills gap problem by automating routine cybersecurity tasks, freeing up human resources to focus on more complex and strategic operations like advanced threat hunting or prioritized cyber resiliency efforts.

The Administration should drive AI cybersecurity adoption through policy integration

- White House leadership is essential for federal adoption. The Administration should embed mandates for Federal agencies to deploy cybersecurity solutions that leverage AI/ML in each key national strategy document, including:
    - National Security Strategy (NSS)
    - National Defense Strategy (NDS)
    - National Military Strategy (NMS)
    - National Cybersecurity Strategy

Reinvigorate the Continuous Diagnostics and Mitigation (CDM) Program

- The CDM program has proven pivotal for agency cybersecurity, but it should evolve to counter emerging threats. A new phase of the CDM should focus on investing in AI-powered solutions, enabling federal agencies to proactively detect and respond to cyber incidents.

Expand Cybersecurity Funding to State & Local Governments

- State and local entities face the same sophisticated cyber threats as federal agencies but often lack resources. Reauthorizing the state/local grant program should fund AI-based cybersecurity solutions for critical infrastructure protection at the SLTT level.

Modernizing standards and controls to encourage adoption of AI-powered cybersecurity

- The best AI capabilities are delivered via the cloud, making cloud security integral to AI security. FedRAMP remains the primary gatekeeper for cloud adoption in government.
- NIST controls must evolve to reflect modern AI-driven security innovations. AI techniques such as pattern recognition, anomaly detection, and behavioral zero trust should be incorporated into cybersecurity controls at the heart of FedRAMP.

2. Issue: Drive AI security across the stack—securing AI data centers, cloud deployments, and models.

Expand AI Security Beyond Model Protection

- While previous policy efforts focused on AI model security, a comprehensive approach must extend to network and cloud environments and AI infrastructure.
- AI security must encompass the full technology stack, including data centers, network, infrastructure, APIs, identities leveraging these assets as well as cloud deployments in addition to the AI models themselves.
- Complete and real-time domain visibility, monitoring, detection and response across all IT and OT protocols is critical for defense and represents the foundation for securing AI systems. Advanced Machine Learning analytics understanding the network or domain are needed in order to detect and contain more sophisticated adversaries.
- OSTP should champion an AI security strategy that extends beyond models, prioritizing network visibility and security whether in the cloud or on premise as well as data center protection, clear communication guardrails, principle of least privilege access controls, and NIST cybersecurity revisions that better include AI use cases for cybersecurity to ensure national resilience.

AI Infrastructure Cybersecurity is Critical

- AI security is not just about the protection of models. Federal AI policy should look more holistically at AI infrastructure writ large—data centers; cloud infrastructure across IaaS, PaaS, & SaaS; data sets; development and deployment workforce, AI infrastructure, including data centers and cloud providers. These should be understood as critical national security assets, requiring AI-driven monitoring and federal hardening efforts akin to CISA's approach to critical infrastructure.
    - The most modern AI is developed in the cloud and deployed through the cloud. Therefore, secure AI deployment means securing cloud deployment.
    - AI workloads rely on high-performing data centers, which are prime targets for cyber threats.

- AI must defend AI—leveraging autonomous cybersecurity solutions to combat new and evolving cyber threats. AI-powered cybersecurity must become standard across federal cybersecurity frameworks. AI should augment zero-trust architectures, making dynamic, behavior-based security adjustments in real time.

3. **Issue: Ensure that any NIST standard on AI governance is interoperable with ISO 42001.**

- The legacy governance approach to AI was not optimized to industry needs.
  - Fractured AI governance weakens U.S. national security and competitiveness.
  - Misaligned standards increase vulnerability by forcing companies to manage conflicting compliance requirements, slowing down AI advancements in defense and intelligence.
- NIST's AI Risk Management Framework (RMF) is a thoughtful standard.
  - The RMF can help U.S. AI companies scale globally by improving alignment with the leading industry-led AI standard, ISO 42001.[1]
  - The Administration should direct NIST to align all AI governance criteria with ISO 42001, the internationally recognized AI governance standard.
- ISO 42001 provides a harmonized framework that strengthens AI supply chains, ensures secure AI deployment across critical sectors, and enhances coordination with allied nations.
  - U.S. companies already comply with ISO 42001, making it the *de facto* global AI governance framework.
  - Aligning AI governance frameworks with a single, clean ISO standard adheres to the President's deregulatory mandate—repealing 10 regulations for each new one.

About Darktrace

Darktrace is a global leader in AI for cybersecurity that keeps organizations ahead of the changing threat landscape every day. Founded in 2013 in Cambridge, UK, Darktrace provides the essential cybersecurity platform to protect organizations from unknown threats using AI that learns from each business in real-time. Its affiliate Darktrace Federal Inc. is on a mission to deliver complete AI-powered solutions to defend U.S. federal government customers from cyber disruptions and ensure mission resilience. Darktrace's platform and services are supported by 2,400+ employees who protect nearly 10,000 customers globally. To learn more, visit http://www.darktrace.com.